Technische Universität Wien

Logic and Computation

course material: slides, official script

"All Star" by Smashmouth but it's expressed in *first-order logic*

Hey now, you're an all-star, get your game on, go play
Hey now, you're a rock star, get the show on, get paid
And all that glitters is gold
Only shooting stars break the mold

Hey now, ∃(x) Px, ∀(x) (Px ⊃ (Rx & Sx))
Hey now, ∃(x) Ax, ∀(x) (Ax ⊃ (Bx & Cx))
∀(x) (Dx ⊃ Ex)
∀(x) (Gx ⊃ Fx)



mash Mouth - All Star - YouTube

Dictionary
Px = x is an all-star
Rx = x is disposed to get their game on
Sx = x is disposed to go play
Ax = x is a rock star
Bx = x is disposed to get the show on
Cx = x is disposed to get paid
Dx = x is glittery
Ex = x is gold
Fx = x is a shooting star
Gx = x has the ability to break the mold

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CHAPTER 1

Proof Theory

Proof theory is concerned with *provability* rather than just truth.

Def. 1 **Deductive System**

A deductive system (calculus) for a logic is a purely syntactic method (i.e. a method that must manipulate only the symbols of the syntax, without using the meaning assigned to them by semantics) for defining the notion of deduction (proof), telling us when a given expression is deducible from a collection of given expressions.

Formally:

$$\Sigma = (L, W, R, Ax)$$

where:

- L is a language
- W is a set of WFF in L
- R is a finite set of n-ary relations on W (inference rules)
- $Ax \subseteq W$ is the (decideable) set of axioms

Def. 2 **Proof**

Let Σ be a deductive system. A proof (or derivation) of P in Σ is a finite tree of WFF rooted in P such that the leafs are formulas in Ax and all the internal nodes are obtained from formulas in Ax and/or (already) derived formulas using a rule in R.

1.1 | Natural Deduction

Natural deduction is a proof calculus in which logical reasoning is expressed by inference rules closely related to the ”natural” way of reasoning, such as introducing temporary assumptions (to introduce an ”if”) or reasoning by cases (to eliminate an ”or”) (in contrast with Hilber-style systems, which use axioms as much as possible).

1.1.1 | Natural Deduction Rules

A key feature of Natural Deduction is the ability to temporarily assume a hypothesis to explore a logical path, and then later cancel (or “discharge”) that hypothesis when introducing certain logical operators. We denote discharged hypotheses by placing them in square brackets, e.g., $[A]$.

Conjunction ($\wedge$)

$$\frac{A \wedge B}{A}\,(\wedge e_1) \qquad \frac{A \wedge B}{B}\,(\wedge e_2) \qquad \frac{A \quad B}{A \wedge B}\,(\wedge i)$$

Disjunction ($\vee$)

$$\frac{A}{A \vee B}\,(\vee i_1) \qquad \frac{B}{A \vee B}\,(\vee i_2) \qquad \frac{A \vee B \qquad \begin{array}{c}[A]\\ \vdots \\ C\end{array} \qquad \begin{array}{c}[B]\\ \vdots \\ C\end{array}}{C}\,(\vee e)$$

($\vee$ e) If you already know $A \vee B$ is true, you have to explore both distinct possibilities to conclude a new fact C. First, you assume A is the true one and derive C; then, you separately assume B is the true one and derive C. Since C naturally follows in either case, C is true regardless of whether A or B triggered it. Both assumptions $[A]$ and $[B]$ are then discharged

Implication ($\rightarrow$)

$$\frac{A \qquad A \to B}{B}\,(\to e) \qquad \frac{\begin{array}{c}[A]\\ \vdots \\ B\end{array}}{A \to B}\,(\to i)$$

(-> i) To prove $A \to B$, we temporarily assume A is true. If, under this assumption, we manage to derive B, we can conclude that $A \to B$ holds unconditionally. At this point, the temporary assumption A is discharged ($[A]$)

Negation ($\neg$) and Falsum ($\bot$)

$$\frac{A \qquad \neg A}{\bot}\,(\neg e) \qquad \frac{\begin{array}{c}[A]\\ \vdots \\ \bot\end{array}}{\neg A}\,(\neg i) \qquad \frac{\bot}{A}\,(\bot e) \qquad \frac{\begin{array}{c}[\neg A]\\ \vdots \\ \bot\end{array}}{A}\,(RAA)$$

($\neg i$) To prove that A is false ($\neg A$), we assume the opposite: that A is true; if calculating from this assumption leads to a contradiction ($\bot$), we can conclude $\neg A$ and discharge the assumption $[A]$

(RAA) (Similar to negation introduction, but used to prove a positive statement A). We assume its negation, $\neg A$: if this leads to a contradiction ($\bot$), we conclude that A must be true, discharging the assumption $[\neg A]$

this rule distinguishes Classical Logic from Intuitionistic Logic !

1.2 | Hilbert Systems

The idea behind Hilbert systems (or Axiomatic systems) is to only use one rule, *Modus Ponens*:

$$\frac{A \qquad A \to B}{B}$$

To these systems introduce many *axioms*, particular WFFs assumed to be true.

Intuitive properties expressed as axioms

By assuming both implication elimination $(\to e)$ and implication introduction $(\to i)$:

$$(\to e) \quad A, A \to B \vdash B$$
$$(\to i) \quad \text{if } A \vdash B \text{ then } \vdash A \to B$$

we obtain the equivalence $A \vdash B \iff \vdash A \to B$.

Consequently, any intuitive deduction property expressed via turnstiles can be transformed into a pure implication statement (axiom schema):

$$A_1, \dots, A_n \vdash A_{n+1} \implies \vdash A_1 \to (A_2 \to (\dots \to (A_n \to A_{n+1}) \dots))$$

The objective of a Hilbert system is to eliminate the rule $(\to i)$.

Def. 3 **Derivation in a Hilbert System**

A *derivation* (or proof) of $\Gamma \vdash A$ in a Hilbert system $\mathcal{H}$ is a tree rooted in A such that:

- its **leaves** are either instances of axiom schemas in $\mathcal{H}$ or formulas belonging to Γ
- its **internal nodes** are obtained from previous nodes or leaves exclusively via *Modus Ponens* (MP)

1.2.1 | Elimination of $(\to i)$ and the Deduction Theorem

Eliminating $(\to i)$ is justified by proving the Deduction Theorem:

Thm. 1 **Deduction Theorem**

$$\text{If } A \vdash B, \text{ then } \vdash A \to B.$$

Proof

Assuming a proof of $A \vdash B$ that uses MP as its only inference rule, we proceed by induction on the number of MP applications, $\#(MP)$:

- **base case ($\#(MP) = 0$):** no applications of MP.
  - case $A = B$: requires introducing the identity axiom schema:

    $$(I) \quad A \to A$$

  - case B is an axiom (or $B \in \Gamma$): requires introducing the weakening axiom schema:

    $$(K) \quad B \to (A \to B)$$

- **inductive step ($\#(MP) = n + 1$):**

  $A \vdash B$ was derived from $A \vdash C \to B$ and $A \vdash C$.

  By the induction hypothesis, we have $\vdash A \to (C \to B)$ and $\vdash A \to C$. To infer $\vdash A \to B$, we add the distribution axiom schema:

  $$(S) \quad (A \to (C \to B)) \to ((A \to C) \to (A \to B))$$

1.2.2 | Axiom Schemas

To represent all standard connectives using Modus Ponens, we can adopt the following axiom schemata:

- **conjunction ($\wedge$):**

$$\begin{aligned}(\wedge e) &\quad A_1 \wedge A_2 \to A_k \quad (k = 1, 2)\\ (\wedge i) &\quad A \to (B \to A \wedge B)\end{aligned}$$

- **disjunction ($\vee$):**

$$\begin{aligned}(\vee i) &\quad A_k \to A_1 \vee A_2 \quad (k = 1, 2)\\ (\vee e) &\quad (A \to C) \to ((B \to C) \to (A \vee B \to C))\end{aligned}$$

- **false ($\bot$) and negation ($\neg$):**

$$\begin{aligned}(\bot e) &\quad \bot \to A\\ (\neg i) &\quad (A \to \bot) \to \neg A\\ (\neg e) &\quad A \to (\neg A \to \bot)\\ \text{(RAA)} &\quad (\neg A \to \bot) \to A\end{aligned}$$

- **implication ($\to$):**

$$\begin{aligned}(K) &\quad B \to (A \to B)\\ (S) &\quad (A \to (C \to B)) \to ((A \to C) \to (A \to B))\end{aligned}$$

1.3 | Equivalence between Natural Deduction and Hilbert Systems

Thm. 2

For any set of formulas Γ and formula P, Natural Deduction (ND) and Hilbert Systems (HS) are equivalent:

$$\Gamma \vdash_{\text{ND}} P \iff \Gamma \vdash_{\text{HS}} P$$

Proof

Direction ($\Longleftarrow$): it is sufficient to show that all axiom schemas of HS are derivable as theorems in ND.

Since Modus Ponens in HS corresponds directly to $(\to\ e)$ in ND, any derivation in HS can be mimicked in ND.

Direction ($\Longrightarrow$): by induction on the depth of the derivation tree for $\Gamma \vdash_{\text{ND}} P$:

- **base case (depth1):** $P \in \Gamma$, which immediately yields $\Gamma \vdash_{\text{HS}} P$
- **inductive step:** assume the claim holds for all derivation trees of depth $\leq n$; we proceed by case analysis on the last inference rule applied in ND:
  - **case** $(\wedge e_1)$**:** The last step is $\frac{A \wedge B}{A}$.

By the inductive hypothesis, $\Gamma \vdash_{\text{HS}} A \wedge B$. Since $(A \wedge B) \to A$ is an axiom schema in HS, applying Modus Ponens yields $\Gamma \vdash_{\text{HS}} A$.

– **case** $(\vee e)$**:** The last step derives C from premises $A \vee B$, $A \vdash C$, and $B \vdash C$.

1 | by the I.H., $\Gamma \vdash_{\text{HS}} A \vee B$, $\Gamma, A \vdash_{\text{HS}} C$, and $\Gamma, B \vdash_{\text{HS}} C$.

2 | applying the Deduction Theorem (DT) to the latter two gives $\Gamma \vdash_{\text{HS}} A \to C$ and $\Gamma \vdash_{\text{HS}} B \to C$.

3 | using the disjunction elimination axiom schema:

$$\vdash_{\text{HS}} (A \to C) \to ((B \to C) \to ((A \vee B) \to C))$$

4 | applying Modus Ponens three times using $\Gamma \vdash_{\text{HS}} A \to C$, $\Gamma \vdash_{\text{HS}} B \to C$, and $\Gamma \vdash_{\text{HS}} A \vee B$ yields $\Gamma \vdash_{\text{HS}} C$

1.4 | Sequent Calculus

Sequent Calculus differs from Natural Deduction and Hilbert Systems in two main ways:

- rules apply not to formulae, but to **sequents** (derivability assertions)
- rules can only *introduce* and not eliminate connectives

A sequent is of the form:

$$A_1, \dots, A_n \vdash B$$

which can be read as "from the premises $A_1, \dots, A_n$ follows the conclusion B".

Some terminology

- **contexts:** the sets of formulas (like Γ or Δ) carried through the rules without being directly modified
- **cut formula:** the formula that is eliminated when applying a cut rule
- **principal formula:** the new formula introduced by a logical rule
- **active formulas:** the sub-formulas in the premises that are combined to build the principal formula

1.4.1 | Sequent Calculus (LJ) Rules

Sequent calculus has four types of rules: axioms, the cut rule, logical and structural rules.

Structural Rules

Structural rules manipulate the context (the sets of formulas surrounding the active formulas) rather than the logical connectives.

$$\frac{\Gamma, B, A, \Delta \vdash C}{\Gamma, A, B, \Delta \vdash C}\ \text{(exchange,l)} \qquad \frac{\Gamma, A, A, \Delta \vdash C}{\Gamma, A, \Delta \vdash C}\ \text{(contraction,l)}$$

$$\frac{\Gamma \vdash B}{\Gamma, A \vdash B}\ \text{(weakening,l)} \qquad \frac{\Gamma \vdash}{\Gamma \vdash B}\ \text{(weakening,r)}$$

(w) you can add extra unused assumptions on the left, or drop to an empty conclusion on the right, without invalidating an already proven sequent

Logical Rules

As mentioned, Sequent Calculus rules only *introduce* connectives, either on the left (manipulating premises) or on the right (manipulating conclusions).

Conjunction ($\wedge$)

$$\frac{\Gamma, A_i \vdash C}{\Gamma, A_1 \wedge A_2 \vdash C}\,(\wedge, l)_i \qquad \frac{\Gamma \vdash A \qquad \Gamma \vdash B}{\Gamma \vdash A \wedge B}\,(\wedge, r)$$

$(\wedge, l)_i$ if C can be derived using just one of the conjuncts A_i alongside the context Γ, it can also be derived when the full conjunction $A_1 \wedge A_2$ is in the premises

$(\wedge, r)$ to prove a conjunction $A \wedge B$ from context Γ, you must independently prove both A and B

Disjunction ($\vee$)

$$\frac{\Gamma, A \vdash C \qquad \Gamma, B \vdash C}{\Gamma, A \vee B \vdash C}\,(\vee, l) \qquad \frac{\Gamma \vdash A_i}{\Gamma \vdash A_1 \vee A_2}\,(\vee, r)_i$$

$(\vee, l)$ to prove C from a disjunction $A \vee B$, you must explore both cases: showing that C follows from assuming A, and separately from assuming B

$(\vee, r)_i$ proving just one of the disjuncts A_i from the context is enough to construct the full disjunction on the right

Implication ($\rightarrow$)

$$\frac{\Gamma, B \vdash C \qquad \Gamma \vdash A}{\Gamma, A \to B \vdash C}\,(\to, l) \qquad \frac{\Gamma, A \vdash B}{\Gamma \vdash A \to B}\,(\to, r)$$

$(\to, l)$ to use an implication $A \to B$ from your premises to prove C, you must first be able to prove its condition A from your context, and then show that the consequence B successfully leads to C

$(\to, r)$ to prove $A \to B$, you temporarily add A to your context Γ and use it to derive B

Negation ($\neg$)

$$\frac{\Gamma \vdash A}{\Gamma, \neg A \vdash}\,(\neg, l) \qquad \frac{\Gamma, A \vdash}{\Gamma \vdash \neg A}\,(\neg, r)$$

$(\neg, l)$ if you can prove A from the context Γ, then adding the premise $\neg A$ creates a contradiction (represented by an empty right side)

$(\neg, r)$ if adding A to your premises leads to an empty conclusion (a contradiction), you can deduce $\neg A$ on the right

The Cut Rule

$$\frac{\Gamma \vdash A \qquad \Gamma, A \vdash B}{\Gamma \vdash B}\ (\text{cut})$$

The cut rule acts as a mechanism for using intermediate lemmas: if you can prove A from your context Γ, and you can use A alongside Γ to prove B, you can deduce B directly from Γ by "cutting" out the intermediate formula A.

> Thm. 3 **Cut-elimination**
>
> Any proof using the (cut) rule can be transformed into a proof without it.

1.4.2 | Soundness, Completeness and Classical Logic

The Sequent Calculus LJ is sound and complete for Intuitionistic Logic.

Furthermore, aside from the CUT rule, the following holds:

> Lemma 1 **Subformula Property**
>
> Everything that appears in the rules' premises appears in their conclusions.

To obtain a calculus for Classical Logic, we need to introduce an axiom or rule that corresponds to Reductio Ad Absurdum.

For example, we could introduce a rule like:

$$\frac{\Gamma, \neg A \vdash}{\Gamma \vdash A}\ (\text{RAA})$$

However, this is problematic because the premise contains more connectives than the conclusion. We do not want this, because the premises must be subformulas of the conclusions to maintain the subformula property.

Alternatively, we could add the Law of Excluded Middle as an axiom ($\vdash A \vee \neg A$), but then we would have to use the CUT rule to make use of it in proofs.

Instead, the solution is to consider sequents that allow **many premises** *and* **many conclusions**:

$$A_1, \dots, A_n \vdash B_1, \dots, B_m$$

In this structure, the formulas on the left side (A_i) are implicitly connected by AND, while the formulas on the right side (B_j) are implicitly connected by OR.

As a consequence of having multiple conclusions, the structural rules must now apply to both the left and the right sides of the turnstile.

1.4.3 | Rules of Sequent Calculus LK'

In LK', structural rules operate symmetrically on both the left and right sides of the sequent.

Structural Rules

$$\frac{\Gamma, B, A, \Delta' \vdash \Delta}{\Gamma, A, B, \Delta' \vdash \Delta}\ (e,l) \qquad \frac{\Gamma \vdash \Delta, B, A, \Delta'}{\Gamma \vdash \Delta, A, B, \Delta'}\ (e,r)$$

$$\frac{\Gamma, A, A \vdash \Delta}{\Gamma, A \vdash \Delta}\,(c,l) \qquad \frac{\Gamma \vdash \Delta, A, A}{\Gamma \vdash \Delta, A}\,(c,r)$$

$$\frac{\Gamma \vdash \Delta}{\Gamma, A \vdash \Delta}\,(w,l) \qquad \frac{\Gamma \vdash \Delta}{\Gamma \vdash \Delta, A}\,(w,r)$$

This calculus also has exchange, contraction and weakening rules.

Logical Rules

Conjunction ($\wedge$)

$$\frac{\Gamma, A_i \vdash \Delta}{\Gamma, A_1 \wedge A_2 \vdash \Delta}\,(\wedge,l)_i \qquad \frac{\Gamma \vdash A, \Delta \qquad \Gamma \vdash B, \Delta}{\Gamma \vdash A \wedge B, \Delta}\,(\wedge,r)$$

$(\wedge,l)_i$ having $A_1 \wedge A_2$ as a premise allows extracting either A_1 or A_2 to prove Δ

Disjunction ($\vee$)

$$\frac{\Gamma, A \vdash \Delta \qquad \Gamma, B \vdash \Delta}{\Gamma, A \vee B \vdash \Delta}\,(\vee,l) \qquad \frac{\Gamma \vdash A_i, \Delta}{\Gamma \vdash A_1 \vee A_2, \Delta}\,(\vee,r)_i$$

$(\vee,r)_i$ proving any one disjunct A_i suffices to establish the disjunction $A_1 \vee A_2$ on the right

Implication ($\rightarrow$)

$$\frac{\Gamma, B \vdash \Delta \qquad \Gamma \vdash A, \Delta}{\Gamma, A \to B \vdash \Delta}\,(\to,l) \qquad \frac{\Gamma, A \vdash B, \Delta}{\Gamma \vdash A \to B, \Delta}\,(\to,r)$$

Negation ($\neg$)

$$\frac{\Gamma \vdash A, \Delta}{\Gamma, \neg A \vdash \Delta}\,(\neg,l) \qquad \frac{\Gamma, A \vdash \Delta}{\Gamma \vdash \neg A, \Delta}\,(\neg,r)$$

$(\neg,l)$ assuming $\neg A$ on the left is equivalent to adding A as an alternative target on the right

$(\neg,r)$ to prove $\neg A$ on the right, move A into the premises on the left

1.4.4 | Formulation Variants: from LK' to LK

In another version of this calculus, LK, the single-formula rules $(\wedge,l)_i$ and $(\vee,r)_i$ are replaced by multi-formula versions that introduce both components at once:

$$\frac{\Gamma, A_1, A_2 \vdash \Delta}{\Gamma, A_1 \wedge A_2 \vdash \Delta}\,(\wedge,l) \qquad \frac{\Gamma \vdash A_1, A_2, \Delta}{\Gamma \vdash A_1 \vee A_2, \Delta}\,(\vee,r)$$

Equivalence of LK' and LK Rule Formulations

By using structural rules (weakening and contraction), the single-formula indexed rules and the multi-formula rules are completely inter-derivable:

- *multi-formula* $(\wedge,l) \implies$ *single-formula* $(\wedge,l)_i$*:*

given $\Gamma, A_1 \vdash \Delta$, apply weakening (w, l) to get $\Gamma, A_1, A_2 \vdash \Delta$, then apply $(\wedge, l)$ to get $\Gamma, A_1 \wedge A_2 \vdash \Delta$

- *single-formula* $(\wedge, l)_i \implies$ *multi-formula* $(\wedge, l)$*:*

  given $\Gamma, A_1, A_2 \vdash \Delta$, apply $(\wedge, l)_1$ to get $\Gamma, A_1 \wedge A_2, A_2 \vdash \Delta$, then $(\wedge, l)_2$ to get $\Gamma, A_1 \wedge A_2, A_1 \wedge A_2 \vdash \Delta$, and finally contraction (c, l) to obtain $\Gamma, A_1 \wedge A_2 \vdash \Delta$:

$$\dfrac{\dfrac{\dfrac{\Gamma, A_1, A_2 \vdash \Delta}{\Gamma, A_1 \wedge A_2, A_2 \vdash \Delta}\,(\wedge, l)_1}{\Gamma, A_1 \wedge A_2, A_1 \wedge A_2 \vdash \Delta}\,(\wedge, l)_2}{\Gamma, A_1 \wedge A_2 \vdash \Delta}\,(c, l)$$

An identical argument holds for $(\vee, r)_i$ and $(\vee, r)$ using (w, r) and (c, r).

Example Derivations in LK

$$\dfrac{\dfrac{\dfrac{A \vdash A}{\vdash A, \neg A}\,(\neg, r)}{\vdash A \vee \neg A, \neg A}\,(\vee, r)_1}{\vdash A \vee \neg A, A \vee \neg A}\,(\vee, r)_2 \qquad \dfrac{\vdash A \vee \neg A, A \vee \neg A}{\vdash A \vee \neg A}\,(c, r)$$

the proof duplicates the target formula on the right via contraction (c, r), enabling both sides of the disjunction to be constructed independently.

$$\dfrac{\dfrac{\dfrac{A \vdash A}{A, \neg A \vdash}\,(\neg, l)}{A \wedge \neg A \vdash}\,(\wedge, l)}{\vdash \neg(A \wedge \neg A)}\,(\neg, r)$$

1.4.5 | Soundness and Completeness of LK

Thm. 4 **Soundness of LK**

If $\Gamma \vdash \Delta$ is derivable, then $\Gamma \vDash \Delta$.

Proof

By induction on the length of the derivation. For each rule, if an interpretation υ validates its premises, then it validates its conclusion. Axioms $A \vdash A$ are trivially valid ($A \vDash A$). Because all rules preserve validity, the end sequent must be valid.

Thm. 5 **Completeness of LK**

If $\Gamma \vDash \Delta$, then $\Gamma \vdash \Delta$ is derivable.

Proof

For each rule, if an interpretation v validates its conclusion, it also validates its premises. This means the logical rules of LK are invertible. Assuming $\Gamma \vDash \Delta$, applying the rules backwards (bottom-up) will eventually yield valid atomic sequents $\Gamma_i \vdash \Delta_i$, which can be derived using axioms and structural rules.

Corollaries of Completeness

- **contraction rules** are redundant
- **weakening rules** are redundant if the axioms are sequents $\Gamma \vdash \Delta$ where $\Gamma \cap \Delta \neq \varnothing$
- the **(cut) rule** is redundant (cut-admissibility)

1.4.6 | Application: Using LK to find CNF

LK can be used to transform a formula P into an equivalent formula P_{CNF} in Conjunctive Normal Form.

If P is not valid, the derivation will yield leaves $A_1, \dots, A_m \vdash B_1, \dots, B_n$ that are not axioms. P is logically equivalent to the conjunction of all such unprovable leaves:

$$\neg A_1 \vee \cdots \vee \neg A_m \vee B_1 \vee \cdots \vee B_n$$

1.5 | Equivalence between Hilbert Systems and Sequent Calculus

Thm. 6

For any set of formulas Γ and formula P:

$$\Gamma \vdash_{\text{HS}} P \iff \Gamma \vdash_{\text{SC}} P$$

Proof

Direction ($\Longrightarrow$): it is sufficient to prove that all HS axioms are theorems in SC, and that Modus Ponens can be simulated. Modus Ponens translates to an application of the $(\rightarrow, l)$ rule combined with the cut rule.

Direction ($\Longleftarrow$): by induction on the depth of the derivation $\Gamma \vdash_{\text{SC}} P$; the logical rules of SC are simulated in HS using the Deduction Theorem (DT) and corresponding HS axioms, similarly to the equivalence proof for Natural Deduction

1.6 | First-Order Logic in Natural Deduction

1.6.1 | Intuitive Properties of Quantifiers

- $(\forall e)$ $\forall x A(x) \vdash A[t/x]$
- $(\forall i)$ if $\Gamma \vdash A[y/x]$ then $\Gamma \vdash \forall x A(x)$ (where y does not appear free in Γ)
- $(\exists i)$ $A[t/x] \vdash \exists x A(x)$
- $(\exists e)$ if $\Gamma, A[y/x] \vdash C$ then $\Gamma, \exists x A(x) \vdash C$ (where y does not appear free in Γ or C)

1.6.2 | Natural Deduction Rules for Quantifiers

$$\frac{A[y/x]}{\forall x A}\,(\forall i) \qquad \frac{\forall x A}{A[t/x]}\,(\forall e)$$

$$\frac{A[t/x]}{\exists x A}\,(\exists i) \qquad \frac{\exists x A \qquad \begin{array}{c}[A[y/x]]\\ \vdots \\ C\end{array}}{C}\,(\exists e)$$

($\forall i$) the eigenvariable y must not appear free in any of the undischarged assumptions on which $A[y/x]$ depends (i.e., any non-canceled leaves of the subtree rooted in $A[y/x]$)

($\exists e$) the eigenvariable y must not appear free in C, nor in any undischarged assumptions of the subtree rooted in C (other than the temporary assumption $A[y/x]$)

1.7 | FOL in Sequent Calculus

To extend the propositional Sequent Calculus LK to First-Order Logic, we add specific rules for the universal ($\forall$) and existential ($\exists$) quantifiers.

1.7.1 | Quantifier Rules in LK

A naive formulation of the quantifier rules can lead to issues with invertibility and infinite proof search branches. To solve this, the rules for introducing quantifiers that instantiate terms ($\forall l$ and $\exists r$) are designed to retain the principal quantified formula in the premises.

$$\frac{\Gamma, A(y) \vdash \Delta}{\Gamma, \exists x A(x) \vdash \Delta}\,(\exists l)^* \qquad \frac{\Gamma \vdash A(t), \exists x A(x), \Delta}{\Gamma \vdash \exists x A(x), \Delta}\,(\exists r)$$

$$\frac{\Gamma, A(t), \forall x A(x) \vdash \Delta}{\Gamma, \forall x A(x) \vdash \Delta}\,(\forall l) \qquad \frac{\Gamma \vdash A(y), \Delta}{\Gamma \vdash \forall x A(x), \Delta}\,(\forall r)^*$$

* in the rules ($\exists l$) and ($\forall r$), the variable y must be an eigenvariable, meaning it does not appear free in $\Gamma \cup \Delta$.

- by keeping $\forall x A(x)$ in the premise of ($\forall l$) and $\exists x A(x)$ in the premise of ($\exists r$), we ensure that the quantifier rules remain invertible during proof search, allowing the same quantified formula to be instantiated multiple times with different terms if necessary

1.7.2 | Proof Search and Completeness

> Algorithm for Proof Search
>
> Given a sequent $\Gamma \vdash \Delta$, the proof search algorithm proceeds as follows:
>
> 1 | try to apply logical (or quantifier) rules backward to the first compound formula of Γ (for odd steps) and Δ (for even steps)
>
> 2 | generated auxiliary formulas are added to the end of the context
>
> 3 | the terms of the language are enumerated. When applying ($\forall l$) or ($\exists r$), select the first term in the enumeration that has not yet been chosen.

The algorithm stops on a branch if it encounters a **success** (the sequent is an Axiom) or a **failure** (the sequent is of the form $\Pi \vdash \Sigma$, where Π and Σ contain only atomic formulas and $\Pi \cap \Sigma = \varnothing$).

Thm. 7 **Completeness of First-Order LK**

$\Gamma \models \Delta \implies \Gamma \vdash \Delta$

Proof

Applying the proof search algorithm yields either a closed proof tree ($\Gamma \vdash \Delta$) or a tree with at least one failure/infinite branch ($\Gamma \not\models \Delta$).

If there is an infinite branch, we extract a *canonical interpretation* (countermodel) out of it. The domain $D_{\mathcal{A}}$ consists of all terms in the language. We assign truth values to atomic formulas based on their presence in the left-hand side of the sequents in the infinite branch.

Using the concept of witnesses (formulas evaluated to false in the premise that trace back to the conclusion), it can be shown that assuming $\Gamma \models \Delta$ alongside an infinite failure branch leads to a contradiction, proving that a valid sequent must produce a finite, closed proof tree.

Immediate Consequences

- First-order classical logic is consistent.
- The LK calculus is complete even without the structural rules of contraction and cut

  cut and contraction are *admissible rules* (adding them does not change the set of provable sequents)

add examples

1.8 | Summary / Cheat Sheet

The Three Systems

- **Natural Deduction (ND):**
  - No axioms. Focuses on *introduction* (i) and *elimination* (e) rules for connectives
  - key element: *assumptions*; you can temporarily assume a formula to explore a logical path, and then discharge it later (e.g., assuming A to prove B, then discharging A to conclude $A \to B$).
- **Hilbert Systems (HS):**
  - Only one rule of inference (*Modus Ponens*). Everything else is handled by Axiom Schemas.
  - Horrible for writing manual proofs, but great for meta-logic and reasoning about the system (only one rule to check in inductive proofs)
- **Sequent Calculus (SC):**
  - Operates on sequents ($\Gamma \vdash \Delta$) rather than bare formulas.
  - Connectives are never eliminated, only introduced (either on the Left or the Right of the turnstile). Also features structural rules (weakening, contraction, exchange) to manage the contexts.

Classical vs. Intuitionistic Logic

How do you upgrade a constructive/intuitionistic system to a Classical one?

- *natural deduction:* add the RAA rule. (Assuming $\neg A$ yields $\bot \implies$ conclude A).
- *sequent calculus:* move from LJ to LK; allow multiple formulas on the right of the turnstile. This implicitly encodes the Law of Excluded Middle, as the right side represents a disjunction (OR).

FOL-specific

When dealing with Quantifiers ($\forall, \exists$), be acareful about the **eigenvariables** !!!

- when generalizing a universal ($\forall i$ in ND, $\forall r$ in SC) or unpacking an existential ($\exists e$ in ND, $\exists l$ in SC), the variable y must be fresh (an eigenvariable). It cannot appear free in your active assumptions/context (*you can't claim something is true for "all x" if you rely on a specific x to prove it.*)
- in LK, the rules ($\forall l$) and ($\exists r$) keep the quantified formula in the premise

  (to prove something like $\exists x A(x)$, you might need to instantiate x multiple times with different terms)

Eigenvariable Horrors

The freshness constraint on eigenvariables is what prevents proof systems from proving logical fallacies. Violating it allows proving false statements like "if one number is even, all numbers are even."

Generalizing over a specific assumption

fallacy: $p(y) \vdash \forall x p(x)$ (*"assuming y is even, every number x is even"*)

- **ND:**

$$\dfrac{\dfrac{P(y)}{\forall x P(x)}\,(\forall i)\ \textbf{[INVALID!]}}{P(y) \to \forall x P(x)}\,(\to i)$$

violation: y is free in the active assumption $P(y)$.

- **LK:**

$$\dfrac{P(y) \vdash P(y)}{P(y) \vdash \forall x P(x)}\,(\forall r)\ \textbf{[INVALID!]}$$

violation: y is free in the left context $\Gamma = \{P(y)\}$.

why: the rule ($\forall i$) requires y to be an *arbitrary* element. But here, y is constrained by the active assumption $P(y)$ (it is a specific y that satisfies P). Semantically, if $P(x)$ means "x is even" over $\mathbb{N}$, assuming $P(2)$ is true, but concluding $\forall x P(x)$ asserts that *every* natural number is even, which is false.

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Witness Out of Scope

fallacy: $\exists x p(x) \vdash \forall x p(x)$ (*"if there exists a prime number, then all numbers are prime"*)

- **Natural Deduction (ND):**

$$\dfrac{\dfrac{\exists x P(x) \qquad [P(y)]^1}{P(y)}\,(\exists e)^1}{\forall x P(x)}\,(\forall i)$$ **[INVALID!]**

violation: y appears free in the conclusion $C = P(y)$ of the $(\exists e)$ step

- **Sequent Calculus (LK):**

$$\dfrac{\dfrac{P(y) \vdash P(y)}{\exists x P(x) \vdash P(y)}\,(\exists l)}{\exists x P(x) \vdash \forall x P(x)}\,(\forall r)$$ **[INVALID!]**

violation: y is free in the right context $\Delta = \{P(y)\}$ when applying $(\exists l)$

why: the variable y inside an existential elimination acts like a local variable, an unknown witness.

In ND, $(\exists e)$ requires that the conclusion C does not contain y free, ensuring y stays inside its local scope. By choosing $C = P(y)$, the witness y "escapes" its scope into the main derivation line. Once outside, it is no longer bound to any assumption, allowing a subsequent $(\forall i)$ step to incorrectly claim that P holds for *all* x.

Rule Equivalences

Goal	Natural Deduction	Sequent Calculus (LK)
use an "if-then" statement	implication elimination $(\to e)$	left implication $(\to l)$
prove an "if-then" statement	implication intro $(\to i)$	right implication $(\to r)$
use a fact	modus ponens / $(\to e)$	the cut rule (admissible)
explore cases of an "or"	disjunction elimination $(\vee e)$	left disjunction $(\vee l)$
prove by contradiction	raa	right side of turnstile (Δ)